

Request for Quotation — Macro Camera Module

Handheld scalp inspection camera · Date: [YYYY-MM-DD] · Reply by: [YYYY-MM-DD]

From	[Company] / [Name] / [Email] / [Phone]
Product	Handheld scalp camera (WiFi, smartphone app) for observing pores, hair shafts and scalp surface. Not a medical diagnostic device.
Request	Camera module with a dedicated macro lens on a VCM for 20 mm working distance (preferred), or an add-on close-up lens for an existing camera (alternative).

1. Sensor (quote either or both)

	Option A	Option B
Sensor	Sony IMX219 — 8 MP, 3280×2464, 1.12 μm	Sony IMX708 — 12 MP, 4608×2592, 1.4 μm
Optical magnification needed	$\approx 0.61\times$	$\approx 1.08\times$
Reference design (thin lens)	EFL ≈ 7.6 mm, sensor-lens ≈ 12 mm	EFL ≈ 10.4 mm, sensor-lens ≈ 21.5 mm
Interface	MIPI CSI-2	MIPI CSI-2

Option A (IMX219) is preferred: shorter module and wider VCM focus range at this magnification.

2. Optical requirements

Item	Target
Working distance	20 mm (module front / window to scalp surface)
Field of view	6 mm horizontal $\pm 5\%$ (vertical follows sensor aspect)
Resolution (object side)	Center ≥ 50 lp/mm (≈ 10 μm); 0.7 field ≥ 35 lp/mm
Depth of field	≥ 0.5 mm at the above resolution
Distortion	$\leq 2\%$
Relative illumination	$\geq 70\%$
Spectrum / coating	420–680 nm (white LED), AR coated
Focus	VCM autofocus, driver IC DW9714 or compatible, same I2C bus as sensor preferred; stroke to cover $\geq \pm 0.5$ mm of working distance
Front window	Glass or sapphire, AR coated, included in the optical design (skin contact, alcohol-wipe resistant)

3. Mechanical

Max. optics outer diameter	[e.g. $\varnothing 14$ mm] — an LED ring must fit around the lens inside a ≈ 35 mm wide handle
FPC / connector	[length / connector type]
Operating temperature	0–40 °C

4. Please provide

1. Optical design summary (layout, EFL, F-number, WD, FOV)	5. NRE / tooling cost and MOQ
2. MTF (object-side lp/mm), depth of field, distortion, rel. illumination	6. Lead time for samples and mass production
3. 2D drawing and 3D model (STEP)	7. Similar products you already make (scalp / skin / intraoral) — existing designs welcome
4. Unit price at samples / 100 / 500 / 1,000 / 5,000 pcs	

5. Planned quantities

Samples	Pilot run	Mass production
[5–10 pcs]	[100–300 pcs]	[1,000 pcs / year]

Alternative (optional): add-on close-up lens

If you quote an add-on lens for an existing IMX219 AF / IMX708 Wide module instead, note that within 20 mm total length an add-on lens cannot fill the sensor with a 6 mm field. Please propose the achievable optical field (expected 12–15 mm with a 6 mm center crop), lens EFL, diameter, spacing and MTF.